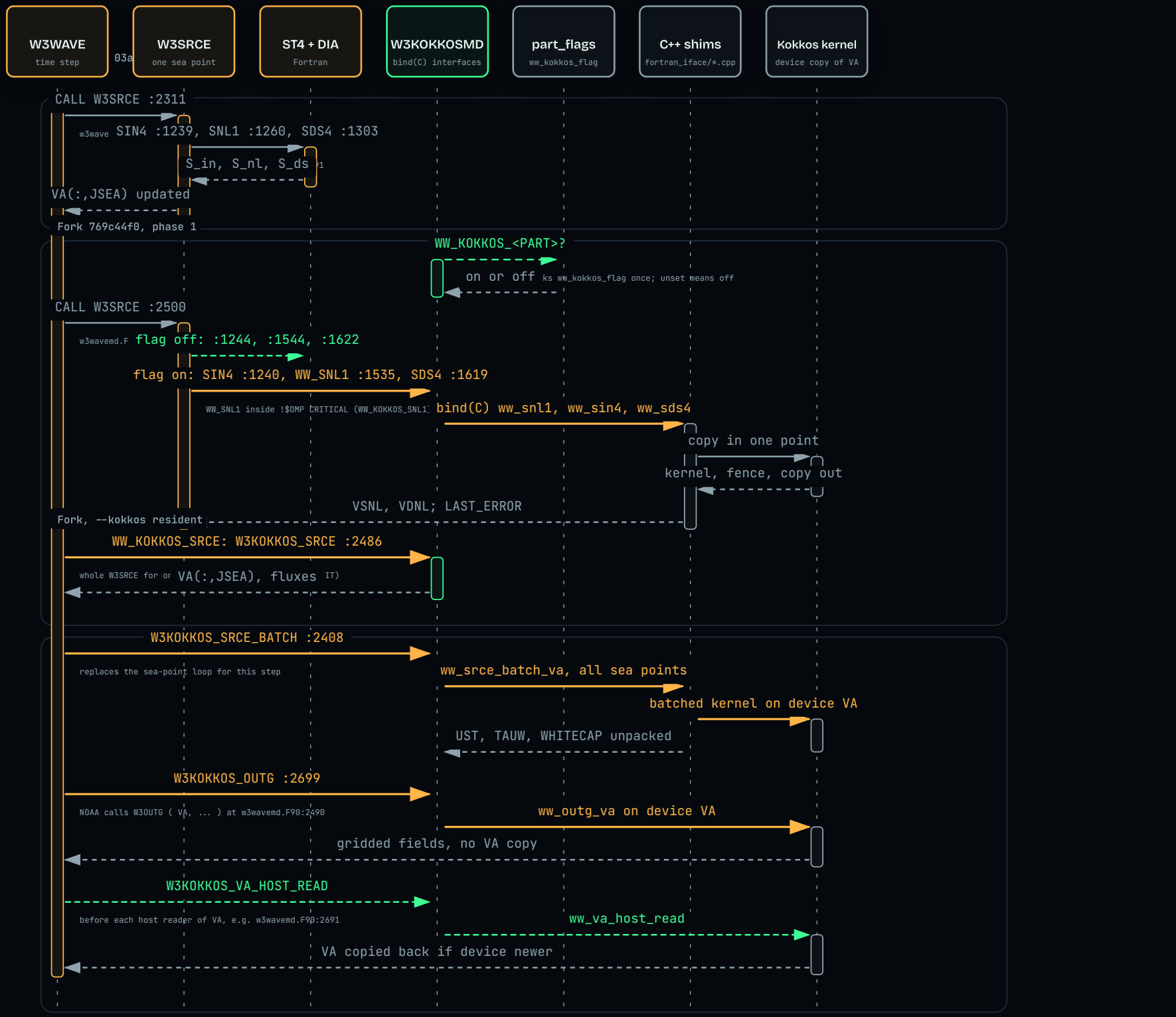

W3WAVE
time step
03a
W3SRCE
one sea point
ST4 + DIA
Fortran
W3KOKKOSMD
bind(C) interfaces
part_flags
ww_kokkos_flag
C++ shims
fortran_iface/*.cpp
Kokkos kernel
device copy of VA
CALL W3SRCE :2311
w3wave SIN4 :1239, SNL1 :1260, SDS4 :1303
S_in, S_nl, S_ds
VA(:,JSEA) updated
Fork 769c44f0, phase 1
WW_KOKKOS_<PART>?
on or off ks ww_kokkos_flag once; unset means off
CALL W3SRCE :2500
w3wavemd.F flag off: :1244, :1544, :1622
flag on: SIN4 :1240, WW_SNL1 :1535, SDS4 :1619
WW_SNL1 inside !$OMP CRITICAL (WW_KOKKOS_SNL1)
bind(C) ww_snl1, ww_sin4, ww_sds4
copy in one point
kernel, fence, copy out
VSNL, VDNL; LAST_ERROR
Fork, --kokkos resident
WW_KOKKOS_SRCE: W3KOKKOS_SRCE :2486
whole W3SRCE for or VA(:,JSEA), fluxes IT)
W3KOKKOS_SRCE_BATCH :2408
replaces the sea-point loop for this step
ww_srce_batch_va, all sea points
batched kernel on device VA
UST, TAUW, WHITECAP unpacked
W3KOKKOS_OUTG :2699
NOAA calls W3OUTG (VA, ...) at w3wavemd.F90:2490
ww_outg_va on device VA
gridded fields, no VA copy
W3KOKKOS_VA_HOST_READ
before each host reader of VA, e.g. w3wavemd.F90:2691
ww_va_host_read
VA copied back if device newer